

Business Analytics – SAS Base

Lecture 1: SAS installation and introduction

Aim of the lecture

- Have SAS University Edition installed in your laptop
- Get you familiar with the user interface of SAS
- Walk you through the power of SAS with an example

Agenda

- **SAS installation**
- SAS user interface
- What can SAS do? An example: Boston Housing Price
- Recap

SAS University Edition is a big move to bring SAS to everyone

- Free, fully functioned (well, internet is needed)
- The installation process can be a bit challenging (so, I am here to help!)
- Let us get SAS University Edition
- Quick Start Guide:



SAS Quick Start.pdf

More information about SAS University edition can be found in the following website

- SAS University Edition:

https://www.sas.com/en_us/software/university-edition.html

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Now let us open SAS and get the first impression

The screenshot displays the SAS Studio web interface. On the left, the 'Server Files and Folders' pane shows a tree structure under 'My Folders' with a folder named 'Boston House'. Inside this folder, several files are listed, including 'bh_data_exploration.sas'. The main editor area on the right has tabs for 'CODE', 'LOG', 'RESULTS', and 'OUTPUT DATA'. The 'CODE' tab is active, showing a SAS program with the following content:

```
1 libname bh '/folders/myfolders/Boston House';  
2  
3 .....  
3 proc import datafile='/folders/myfolders/Boston House/boston_house_train.csv'  
4 dbms = csv  
5 out= bh.boston_house_train;  
6 getnames= yes;  
7 run;  
8  
9
```

The status bar at the bottom indicates the current file path as '/folders/myfolders/Boston House/bh_data_exploration.sas', the cursor position as 'Line 5, Column 28', and the encoding as 'UTF-8'. On the far right of the status bar, it shows 'Messages: 6' and 'User: sasdemo'.

In the long term, it is more efficient for you to be familiar with some keyboard shortcuts

Keyboard shortcut	What can it do?
F3	Run the (selected) script
Ctrl + /	Comment the selected script
Ctrl + Shift + /	Uncomment the selected script
Ctrl + Shift + U	Convert to upper case
Ctrl + Shift + L	Convert to lower case

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The aim of the example is to show what you could do with SAS **after** this class

- Do not get overwhelmed
 - Script is provided and you just need to run it with me
 - And yes, over the course you will gradually crack this script (and write even better ones)
- Growth mindset is important for your learning
- Ready, let's go!

Boston housing price is a very good open dataset to practice your SAS skill

Data Extraction

- Read the dataset into SAS

Data Manipulation

- Use Data functions to deal with missing field, derived field, date, character, sort the data, etc.

Statistical analysis

- Data exploration and analysis (distribution, correlation, etc.)

Automation

- Out of scope for this course

Recap: Now you have SAS installed and know what you can do with SAS!

- SAS University Edition provides opportunity for everyone to learn SAS
- The key windows are code, log, result, and output data
- SAS is a very comprehensive tool that can be leveraged in the whole analysis cycle

*Thank
you*

